Amy Wang

Contact Information

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Professional Summary

Computational scientist integrating biophysical frameworks with advanced AI/ML to accelerate therapeutic protein design. Dual expertise in wet-lab and computational methods enables end-to-end impact in biotechnology.

Research Experience

Genentech, Prescient Design — South San Francisco, CA
Senior ML Scientist, Large Molecule Drug Discovery — Mar 2023 – Present

- Pioneered the application of a known design strategy to a novel area of antibody engineering, which was adopted across 3+ portfolio programs to enable multi-property optimization and validated through cross-departmental collaboration
- Filed 2 provisional patents on (1) generative AI-powered developability optimization and (2) interpretable ML affinity prediction, enabling therapeutic protein design and structural insights from sequence data
- Developed and deployed AI- and rule-based frameworks for antibody engineering, integrated across 10+ programs to accelerate molecule design decisions
- Directed and executed technical research in an external collaboration in studies on dataset scaling and affinity maturation using a Bayesian optimization framework
- Established standards for model adoption by benchmarking computational methods and curating pipeline-relevant ML training datasets, which strengthened collaboration with experimental teams
- Communicated impactful findings in 1 journal submission, 2 conference proceedings, 3 workshop papers.

Microsoft Research New England — Cambridge, MA (*Virtual*)
BioML Research Intern — Jun-Sept 2022
Advisors: Kevin Yang, Ava Amini & Alex Lu

- Conceptualized and implemented integration of biophysical features into graph neural networks for protein property prediction
- Presented findings at *NeurIPS Machine Learning for Structural Biology* Workshop showing Rosetta computed features improve out-of-distribution prediction on two benchmark tasks

Stanford University — Stanford, CA
Graduate Student Researcher — Apr 2019-March 2023
Advisors: William Weis & Alexander Dunn

- Characterized key structural determinants of force-dependent cell adhesion mechanisms using single molecule force spectroscopy (SMFS) and quantitative biochemical assays
- Developed theoretical framework explaining how cooperative molecular interactions generate catch bonds, advancing understanding of adhesion biology mechanics in physiological shear conditions
- Devised an pipeline which increased data collection and analysis throughput by 10x
- Built a custom single-channel epi-fluorescent microscope

Stanford University — Stanford, CA
Rotation Student — Jan-Apr 2019
Advisor: Ron Dror

- Conducted molecular dynamics simulations of membrane protein ion transport, contributing to findings published in Nature

Education

Stanford University, Stanford, CA 2018-2023
Ph.D., Chemical Engineering
Thesis: Mechanisms governing the force-dependent interactions between αE-catenin and F-actin
Advisors:
William Weis (Depts. Structural Biology, Molecular and Cellular Physiology)
Alexander Dunn (Dept. Chemical Engineering, Biophysics program)

Massachusetts Institute of Technology, Cambridge, MA 2014-2018
Bachelor of Science in Chemical Engineering, Minor in Polymers and Soft Matter

Publications and Preprints

N.C. Frey, I. Hötzel, S.D. Stanton, R. Kelly, R.G. Alberstein, ..., **A. Wang**, ..., A. Regev, Y. Wu, K. Cho, R. Bonneau, V. Gligorijević. Lab-in-the-loop therapeutic antibody design with deep learning. *bioRxiv* 2025. [Link]

A.A. Ismail, T. Oikarinen, **A. Wang**, J. Adebayo, S.D. Stanton, H.C. Bravo, K. Cho, N.C. Frey. Concept Bottleneck Language Models For Protein Design. *International Conference on Learning Representations (ICLR)* 2025. [Link]

C. Wang, M. Uehara, Y. He, **A. Wang**, A. Lal, T. Jaakola, S. Levine, A. Regev, H. Wang, T. Biancalani. Fine-tuning discrete diffusion models via reward optimization with applications to DNA and protein design. *International Conference on Learning Representations (ICLR)* 2025. [Link]

NA Bax*, **A Wang***, DL Huang, S Pokutta, AR Dunn, WI Weis. Multi-level force-dependent allosteric enhancement of αE-catenin binding to F-actin by vinculin. *Journal of Molecular Biology* 2023. [Link]

A Wang, AR Dunn, WI Weis. Mechanism of the cadherin-catenin F-actin catch bond interaction. *eLife* 2022. [Link]

S Bose, LR Volpatti*, D Thiono*, V Yesilyurt, C McGladrigan, Y Tang, A Facklam, **A Wang**, S Jhunjhunwala, O Veiseh, J Hollister-Lock, C Bhattacharya, GC Weir, DL Greiner, R Langer, DG Anderson. A retrievable implant for the long-term encapsulation and survival of therapeutic xenogeneic cells. *Nat. Biomed. Eng.* 2020. [Link]

TA Chew*, BJ Orlando*, J Zhang*, NR Latorraca, **A Wang**, SA Hollingsworth, DH Chen, RO Dror, M Liao, L Feng. Structure and mechanism of the cation–chloride cotransporter NKCC1. *Nature* 2019 572:488–492. [Link]

A Huang, JM Paloni, **A Wang**, AC Obermeyer, HV Sureka, H Yao, BD Olsen. Predicting protein–polymer block copolymer self-assembly from protein properties. *Biomacromolecules* 2019, 20, 10, 3713-3723. [Link]

S Chopra, N Bertrand, J Lim, **A Wang**, O Farokhzad, R Karnik. Design of insulin-loaded nanoparticles enabled by multistep control of nanoprecipitation and zinc chelation. *ACS Applied Materials & Interfaces* 2017 9 (13), 11440-11450. [Link]

Workshop Papers

A Wang, Z. Sang, S.D. Stanton, J.L. Hofmann, [...], N.C. Frey, A.M. Watkins, F. Seeger. A guided design framework for the optimization of therapeutic-like antibodies. *Generative and Experimental Perspectives for Biomolecular Design, ICLR* 2025.

P. Rao, H. Isaacson, J.L. Hofmann, D. Davidson, **A. Wang**, A.M. Watkins, R. Bonneau, S. Izadi, J.H. Lee. SurfProp: A surface-based property prediction framework for antibody developability and screening *GenBio Workshop, ICML* 2025.

Z. Ma, D. Davidson, **A Wang**, N. Frey, F. Seeger. The effects of structural conditioning on antibody inverse Folding. *Women in ML Workshop, NeurIPS* 2023.

A Wang, AP Amini, AX Lu, KK Yang. Learning from physics-based features improves protein property prediction. *Machine Learning in Structural Biology Workshop, NeurIPS* 2022. [Link]

Other Work Experience

Stanford Office of Technology Licensing — Stanford, CA
Intern — Oct 2021-Oct 2022
- Evaluated invention disclosures, analyzed patent landscape, and prepared marketing abstracts

Merck Research Labs — West Point, PA
Discovery Pharmaceutical Sciences Research Intern — Jun-Sept 2017
Advisor: Lauren Austin
- Synthesized and characterized nanoparticles for RNA-based vaccine technologies
- Quantified kinetics of peptide and nucleic acid degradation *in-vitro*

Goldman Sachs — New York, NY
Structured Credit Trading Analyst Intern — Jan 2016
- Performed credit risk analyses and built financial valuation models

Massachusetts Institute of Technology — Cambridge, MA
Undergraduate Researcher — Jan 2016-Jun 2018
Advisor: Bradley Olsen
- Synthesized protein-polymer materials and quantified kinetics of functionalized proteins

Koch Institute of Integrative Cancer Research — Cambridge, MA
Undergraduate Researcher — 2014-2016
Advisors: Robert Langer & Rohit Karnik
- Synthesized and characterized nanoparticles to improve insulin encapsulation
- Mentored an undergraduate researcher in biological lab techniques

Honors and Awards

University Nominee for the Schmidt Science Fellows *8 Stanford students nominated*	2022
Valuation of Public Companies in the Life Sciences Pitch Competition - First Place	2020
Stanford Graduate Fellowship in Science & Engineering	2018-2023
NSF Graduate Research Fellowship	2018-2023
ChemH Chemistry/Biology Interface Predoctoral Training Program	2018-2023
Tau Beta Pi Honors Society	2017
MIT Chemical Engineering Departmental BP Academic Achievement Award	2016

Service and Leadership

Stanford Graduate Life Office, Head Community Associate	2021-2023
Stanford Molecular & Cellular Physiology DEI Committee Liaison	2020-2022
Stanford Graduate Life Office, Community Associate	2019-2021

Bay Area Graduate Pathways to STEM Mentor & Panel Speaker	2018-2020
MIT AIChE Vice President, External Relations	2016-2018
MIT Undergraduate Advising & Academic Programming Steering Committee	2015-2017

Skills

Machine Learning
Bayesian optimization, generative modeling, parameter estimation, Monte Carlo simulation

Biophysical Modeling
PyMOL, PyRosetta, AMBER molecular dynamics

Experimental
Biophysical: Optical tweezers, SPR
Molecular biology: Mammalian and bacterial cell culture, FPLC, cloning, western blots, actin pelleting assays